

Algorithm:

1. Start.
2. Read the student's first name, middle name, and last name.
3. Display a menu with eight options.
4. Read the user's choice.
5. Use switch-case to perform the selected operation using a separate function.
6. For Combine Names, join first, middle, and last names using strcpy() and strcat().
7. For Find Length, use strlen() to determine the length of the full name.
8. For Copy Name, copy the full name using strcpy().
9. For Compare Names, compare first and last names using strcmp().
10. For Reverse Name Order, join last, middle, and first names.
11. For Count Characters, check every character and count vowels, consonants, digits, and spaces.
12. For Search Word, read a word and search for it using strstr().
13. Repeat the menu until the user selects Exit.
14. Stop.

C Program:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
/* Function declarations */
```

```
void combineNames(const char firstName[], const char middleName[],  
                 const char lastName[], char fullName[]);
```

```
void findLength(const char fullName[], size_t *length);
```

```
void copyName(const char source[], char destination[]);
```

```
void compareNames(const char firstName[], const char lastName[],  
    int *comparisonResult);
```

```
void reverseNameOrder(const char firstName[], const char middleName[],  
    const char lastName[], char reverseName[]);
```

```
void countCharacters(const char fullName[], int *vowels,  
    int *consonants, int *digits, int *spaces);
```

```
void searchWord(const char fullName[], const char word[],  
    int *searchResult);
```

```
int main()  
{  
    char firstName[50], middleName[50], lastName[50];  
    char fullName[150], copiedName[150], reverseName[150];  
    char word[50];  
  
    int choice;  
    int comparisonResult, searchResult;  
    int vowels, consonants, digits, spaces;  
    size_t length;  
  
    /* Read student name */  
    printf("Enter first name: ");
```

```
scanf("%49s", firstName);
```

```
printf("Enter middle name: ");
```

```
scanf("%49s", middleName);
```

```
printf("Enter last name: ");
```

```
scanf("%49s", lastName);
```

```
/* Create full name initially */
```

```
combineNames(firstName, middleName, lastName, fullName);
```

```
do
```

```
{
```

```
    printf("\n----- STUDENT NAME OPERATIONS ----- \n");
```

```
    printf("1. Combine names and display full name\n");
```

```
    printf("2. Find length of full name\n");
```

```
    printf("3. Copy full name\n");
```

```
    printf("4. Compare first name and last name\n");
```

```
    printf("5. Display name in reverse order\n");
```

```
    printf("6. Count vowels, consonants, digits and spaces\n");
```

```
    printf("7. Search for a word in full name\n");
```

```
    printf("8. Exit\n");
```

```
    printf("Enter your choice: ");
```

```
    scanf("%d", &choice);
```

```
    switch (choice)
```

```
{
```

case 1:

```
combineNames(firstName, middleName, lastName, fullName);
```

```
printf("Full Name: %s\n", fullName);
```

```
break;
```

case 2:

```
findLength(fullName, &length);
```

```
printf("Length of full name: %zu\n", length);
```

```
break;
```

case 3:

```
copyName(fullName, copiedName);
```

```
printf("Copied Name: %s\n", copiedName);
```

```
break;
```

case 4:

```
compareNames(firstName, lastName, &comparisonResult);
```

```
if (comparisonResult == 0)
```

```
    printf("First name and last name are equal.\n");
```

```
else
```

```
    printf("First name and last name are different.\n");
```

```
break;
```

case 5:

```
reverseNameOrder(firstName, middleName,  
                  lastName, reverseName);
```

```
printf("Name in reverse order: %s\n", reverseName);
```

```
break;
```

case 6:

```
countCharacters(fullName, &vowels, &consonants,  
                &digits, &spaces);
```

```
printf("Vowels: %d\n", vowels);
```

```
printf("Consonants: %d\n", consonants);
```

```
printf("Digits: %d\n", digits);
```

```
printf("Spaces: %d\n", spaces);
```

```
break;
```

case 7:

```
printf("Enter word to search: ");
```

```
scanf("%49s", word);
```

```

        searchWord(fullName, word, &searchResult);

        if (searchResult == 1)
            printf("Word found in the full name.\n");
        else
            printf("Word not found in the full name.\n");

        break;

    case 8:
        printf("Exiting program...\n");
        break;

    default:
        printf("Invalid choice! Please enter 1 to 8.\n");
    }

} while (choice != 8);

return 0;
}

/* 1. Combine first, middle and last names */
void combineNames(const char firstName[], const char middleName[],
                 const char lastName[], char fullName[])

```

```
{  
    strcpy(fullName, firstName);  
  
    strcat(fullName, " ");  
    strcat(fullName, middleName);  
  
    strcat(fullName, " ");  
    strcat(fullName, lastName);  
}
```

/* 2. Find length of full name */

```
void findLength(const char fullName[], size_t *length)  
{  
    *length = strlen(fullName);  
}
```

/* 3. Copy full name */

```
void copyName(const char source[], char destination[])  
{  
    strcpy(destination, source);  
}
```

/* 4. Compare first and last names */

```
void compareNames(const char firstName[], const char lastName[],  
                  int *comparisonResult)
```

```
{  
    *comparisonResult = strcmp(firstName, lastName);  
}
```

/* 5. Reverse the order of names */

```
void reverseNameOrder(const char firstName[], const char middleName[],  
                     const char lastName[], char reverseName[])  
{  
    strcpy(reverseName, lastName);  
  
    strcat(reverseName, " ");  
    strcat(reverseName, middleName);  
  
    strcat(reverseName, " ");  
    strcat(reverseName, firstName);  
}
```

/* 6. Count vowels, consonants, digits and spaces */

```
void countCharacters(const char fullName[], int *vowels,  
                   int *consonants, int *digits, int *spaces)  
{  
    int i;  
    char ch;  
  
    *vowels = 0;  
    *consonants = 0;
```

```
*digits = 0;
```

```
*spaces = 0;
```

```
for (i = 0; i < strlen(fullName); i++)
```

```
{
```

```
    ch = tolower(fullName[i]);
```

```
    if (isalpha(ch))
```

```
    {
```

```
        if (ch == 'a' || ch == 'e' || ch == 'i' ||
```

```
            ch == 'o' || ch == 'u')
```

```
        {
```

```
            (*vowels)++;
```

```
        }
```

```
    else
```

```
    {
```

```
        (*consonants)++;
```

```
    }
```

```
}
```

```
else if (isdigit(ch))
```

```
{
```

```
    (*digits)++;
```

```
}
```

```
else if (isspace(ch))
```

```
{
```

```
    (*spaces)++;
```

```
}
```

```
}
```

```
}
```

```
/* 7. Search for a word in full name */
```

```
void searchWord(const char fullName[], const char word[],
```

```
    int *searchResult)
```

```
{
```

```
    if (strstr(fullName, word) != NULL)
```

```
        *searchResult = 1;
```

```
    else
```

```
        *searchResult = 0;
```

```
}
```

TEST CASE TABLE:

test	operations	input	Expected output
1	Combine names	Aman raj sharma	Aman raj sharma
2	Find length	Aman raj sharma	15
3	Copy name	Aman raj sharma	Aman raj sharma
4	Compare	First = Aman, last = sharma	Names are different
5	Reverse order	Aman raj sharma	Sharma Raj aman
6	Count characters	Aman raj sharma	Vowels = 5, Consonants = 8, Digits = 0, Spaces = 2
7	Search word	Search raj	Word found
8	Search word	Search kumar	Word not found
9	exit	Choice = 8	Program exits